

Manual Test Checklist

ECITY — Mobile Shop Management Web App

Version 1.1 · 2026

What a person still has to judge, module by module

Companion documents: PRD, Module Breakdown, Engineering Design, Deployment Guide, Database Guide

How to Use This

Automated tests cover the mechanics — 321 unit and integration tests, 676 browser tests across four screen sizes. This checklist covers what a person still has to judge: does it *feel* right, does the wording make sense, does the shop's actual workflow survive contact with the screen.

Work through a module's section after it is built, and again before go-live. Tick as you go. Anything that fails is a bug, not a note.

One section per module. M1–M14 append to this file as they ship.

Before you start

```
docker compose up -d      # or open Postgres.app
npm run db:migrate
npm run db:seed           # resets the three test accounts
npm run dev               # http://localhost:3000
```

To return to a known-clean state at any point, re-run `npm run db:seed`. It upserts, so it is safe to repeat.

Test accounts

Email	Password	Role	Branch scope
admin@ecity.local	ChangeMe!2026	Admin / Owner	All branches (14 permissions)
manager@ecity.local	ChangeMe!2026	Branch Manager	MAIN only (7 permissions)
staff@ecity.local	ChangeMe!2026	Staff	MAIN only (3 permissions)

ChangeMe!2026 is only the fallback the seed uses when `SEED_PASSWORD` is not set — which is the normal case on your own machine. **On staging or any deployed server the password is whatever `SEED_PASSWORD` was set to** in that server's `.env`; the accounts above are otherwise identical. Read it back with `grep SEED_PASSWORD ~/app/.env`.

Use a **private/incognito window** to hold a second session, so you can be two users at once without signing out.

These accounts must never exist in production. They are flagged `mustChangePassword` and are created only by the seed.

M0 — Foundations

Delivers. PRD FR-1.1 – FR-1.5, FR-31.1, plus branch context and the authorisation layer everything else depends on.

Signing in as the admin exercises perhaps a tenth of this module. The interesting part is what the *other two* users cannot do.

1. Authentication

Start signed out, at `/login`.

- [] Wrong password for a real account → *"Email or password is incorrect."*
- [] A made-up email with any password → **the identical message**. It must not reveal whether the account exists
- [] Empty email or password → inline validation, no request sent

- [] Visit `/settings/users` directly while signed out → redirected to `/login`
- [] Correct credentials → lands on `/dashboard`
- [] Avatar menu → **Sign out** → back at `/login`
- [] After signing out, the browser Back button does **not** restore the app
- [] Sign in again in a second browser → both sessions work independently

Password reset

Email is not wired up until M14, so the reset link is shown on screen in development.

- [] `/forgot-password` → enter `admin@ecity.local` → confirmation appears
- [] A **yellow development box** shows the reset link
- [] Enter an address that does not exist → **the same confirmation**, no hint that the account is unknown
- [] Follow the link, set a new password → redirected to `/login`
- [] The **old** password no longer works
- [] The **new** password works
- [] Open the same reset link a second time → refused as already used
- [] Reset the password back to `ChangeMe!2026`, or re-run `npm run db:seed`

Logout

- [] Get the password wrong 8 times in a row → *"Too many failed attempts."*
- [] Wait 15 minutes, or `npm run db:seed`, to clear it

2. Branch scoping

The heart of M0. Sign in as each user and open the branch switcher in the header.

- [] **admin** — sees *All branches (consolidated)*, MAIN and NORTH
- [] **manager** — sees **MAIN only**. No NORTH, and no "All branches"
- [] **staff** — sees MAIN only
- [] As admin, switch between branches → the dashboard's *Branch context* card updates (active branch, accessible count, consolidated yes/no)
- [] The chosen branch survives a page reload
- [] On a phone-width window the switcher shows the branch **code**, not the full name, and the header still fits

3. Permissions

Check what is **missing**, not what is present.

User	Sidebar should show
admin	Dashboard, Users, Roles, Audit log
manager	Dashboard, Users, Audit log — no Roles
staff	Dashboard only

- [] Each user's sidebar matches the table
- [] As **staff**, type `/settings/roles` into the address bar → redirected, not shown
- [] As **staff**, type `/settings/users` → redirected
- [] As **manager**, `/settings/roles` → redirected
- [] The dashboard's *Your permissions* card lists 14 / 7 / 3 entries respectively

4. Server-side authorisation

A hidden menu item is not a control. This step proves the server refuses the request, not just that the button was hidden.

Sign in as **staff** in the browser, then copy the session cookie from DevTools → Application → Cookies → `ecity_session`, and run:

```
COOKIE='ecity_session=PASTE_VALUE_HERE'
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/users -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/audit -H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/roles -H "Cookie: $COOKIE"
```

- [] All three return **403**, not data
 - [] The body reads `{"error":"...", "code":"FORBIDDEN"}`
 - [] With no cookie at all → **401**
 - [] As **manager**, forcing another branch is refused:
- ```
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/session/branch \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"branchId":2}'
```
- [ ] Returns **403** — branch NORTH is not theirs
  - [ ] `{"branchId":null}` (the consolidated view) also returns **403**

## 5. User management

As **admin**, Users → Add user.

- [ ] Password under 10 characters → rejected inline
- [ ] An email that already exists → *"A user with this email already exists."*
- [ ] No role chosen → *"Choose a role."*
- [ ] Create a Staff user assigned to **NORTH** only
- [ ] They appear in the user list with the right role and branch badge
- [ ] Sign in as the new user → branch switcher shows **NORTH only**
- [ ] Their first sign-in is flagged "must change password"

### Roles

As **admin**, Settings → Roles.

- [ ] Select **Staff**, tick *View the audit log*, Save
- [ ] Sign in as staff → **Audit log now appears** in their sidebar
- [ ] Any staff session open elsewhere was **signed out** when the role changed, rather than keeping stale permissions until next login

- ☐ Untick it and Save again
- ☐ Try to remove *Create and edit users* from **Admin** → refused, so the business cannot lock itself out
- ☐ System roles cannot be deleted

## 6. Audit log

---

As **admin**, Settings → Audit log.

- ☐ Everything done above appears: `LOGIN`, `LOGIN_FAILED`, `LOGOUT`, `CREATE`, `PERMISSION_CHANGED`, `PASSWORD_RESET_COMPLETED`
- ☐ Each row shows who, what, when
- ☐ Expanding a "*n field(s)*" row shows old → new values
- ☐ A password hash **never** appears in a changes payload
- ☐ Paging works if there are more than 50 entries

The log is append-only, enforced by the database rather than by code:

```
psql ecity -c "update audit_log set summary='tampered' where id=1;"
psql ecity -c "delete from audit_log where id=1;"
```

- ☐ Both fail with `audit_log is append-only`

## 7. Responsive layout

---

Resize the window, or use DevTools device mode. Check at ~375px, ~810px and full width.

- ☐ Below ~768px the sidebar is replaced by a **hamburger** top-left
- ☐ Tapping it opens a drawer; choosing a destination closes it
- ☐ On **Users**, the table becomes **one card per user**
- ☐ On **Audit log**, entries become cards
- ☐ Nothing scrolls sideways at any width
- ☐ Buttons and links are comfortably tappable with a thumb
- ☐ The **Roles** editor is usable — role list becomes a horizontal strip
- ☐ Forms stack, with full-width buttons

## 8. Theme

---

- ☐ Primary colour is **black** throughout (buttons, active nav item)
- ☐ Set `APP_THEME=navy` in `.env`, restart → the whole app is blue, with no other visual change
- ☐ An invalid value such as `APP_THEME=nonsense` → falls back to black rather than breaking
- ☐ Set it back to `onyx`

## What M0 deliberately does not include

---

Do not raise these as bugs:

- No products, inventory, IMEIs, customers or suppliers — those are M1 and M2
- The dashboard shows branch context and permissions, **not** sales figures. The real dashboard is M10
- No branch create/edit screen — M1. The two seeded branches are fixed
- Password reset emails print to screen instead of sending — M14
- Notifications, reports, exports — later modules

## M1 — Master Data

**Delivers.** PRD FR-2.1 – FR-2.4, FR-3.1, FR-3.2, FR-3.8, FR-5.10, FR-6.6, FR-10.1.

The shop gets configured and the people it trades with exist. Nothing here moves stock or money yet — these are the pickers and directories the rest of the system draws on.

**Customer** = someone who buys from the shop. **Supplier** = someone the shop buys from. Both are shared across every branch.

### 1. Business settings

Sign in as **admin**, go to **Settings** → **Business**.

- [ ] Four tabs: Profile, Tax, Payments, Expenses
- [ ] On a phone the tabs scroll sideways rather than wrapping

#### Profile

- [ ] Edit the business name, save, reload — the change persisted
- [ ] Enter a malformed GST number → *"Enter a valid 15-character GSTIN."*
- [ ] Enter a valid one (29ABCDE1234F1Z5) → accepted, stored upper-case
- [ ] Currency is fixed at INR and cannot be edited
- [ ] Toggle **Prices include tax** → the explanation below it changes
- [ ] Save, reload → the toggle held its new position

#### Tax

- [ ] Five GST slabs are listed: 0, 5, 12, 18, 28%
- [ ] **GST 18%** is marked Default
- [ ] Rates display as 18.00%
- [ ] Add a rate named **GST 3%** at 3 → appears in the list
- [ ] Add one with the same name → refused as a duplicate
- [ ] Deactivate a non-default rate → shows Inactive
- [ ] Try to deactivate the **default** rate → refused, with the reason

#### Payments

- [ ] Cash, UPI, Card, Bank Transfer are listed with their type badge
- [ ] Cash carries a **Cash drawer** badge; the others do not
- [ ] Deactivate UPI → shows Inactive; reactivate it

- [ ] Try to deactivate **Cash** → refused: at least one cash method must stay active

## Expenses

- [ ] Seven categories: Rent, Electricity, Salaries, Transport, Packaging, Repairs, Miscellaneous
- [ ] Add one → appears immediately
- [ ] Add the same name again → refused

## 2. Branches

---

**Settings** → **Branches** as admin.

- [ ] MAIN and NORTH are listed with manager and user counts
- [ ] **Add branch** — code **MGROAD**, name **MG Road** → created and listed
- [ ] The code is forced to upper case even if typed lower
- [ ] Add another with code **MGROAD** → *"Branch code MGROAD is already in use."*
- [ ] A code containing a space or ! is rejected inline
- [ ] Open the new branch, change its name, save → the list shows the new name
- [ ] Assign a manager from the dropdown → shows in the Manager column
- [ ] Leave the manager empty → **saves without error** (*this failed silently before — see §5*)

### Deactivation — the rule that matters (FR-3.8)

- [ ] Deactivate **MG Road** → badge turns Inactive, **the row stays**
- [ ] Open the header **branch switcher** → MG Road is **gone**
- [ ] Reactivate it → it comes back in the switcher
- [ ] Deactivate every branch but one, then try the last → refused: at least one branch must stay active

## 3. Customers and suppliers

---

**Customers** in the sidebar.

- [ ] The list is empty, or holds only what you have created
- [ ] **Add customer** with a name only → saves. A walk-in has nothing else
- [ ] Add one with phone **9876543210**
- [ ] Add a second with the **same phone** → *"Another customer (...) already uses this phone number."*
- [ ] Give a **supplier** that same phone → **allowed**, they are different directories
- [ ] Edit a customer, keep their phone unchanged, save → no false duplicate
- [ ] Enter **not-a-phone** → rejected inline
- [ ] Enter **MiXeD@Example.COM** → stored lower-case (check in Studio)

### Search and status

- [ ] Search by part of a name → matches
- [ ] Search by phone → matches
- [ ] Search by email → matches

- [ ] Search by GST number → matches
- [ ] **Deactivate** a customer → disappears from the default list
- [ ] Click **Show inactive** → they reappear, badged Inactive
- [ ] Reactivate → back to normal

Repeat the same pass on **Suppliers**. Suppliers additionally have a **Company** field; customers do not.

## 4. Permissions

| User    | Should see in the sidebar                                                       |
|---------|---------------------------------------------------------------------------------|
| admin   | Customers, Suppliers, Business, Branches, Users, Roles, Audit log               |
| manager | Customers, Suppliers, Business, Users, Audit log — <b>no Branches, no Roles</b> |
| staff   | Customers, Suppliers — <b>no Settings group at all</b>                          |

- [ ] Each user's sidebar matches
- [ ] As **staff**, open </settings/business> directly → redirected
- [ ] As **staff**, open </settings/branches> directly → redirected
- [ ] As **staff**, open </customers/new> directly → redirected (they may view customers, not create them)
- [ ] As **manager**, </settings/branches> → redirected. Managers run a branch; they do not create them

At the API, signed in as staff (see M0 §4 for how to get the cookie):

```
curl -s -o /dev/null -w '%{http_code}\n' localhost:3000/api/branches?manage=1 \
-H "Cookie: $COOKIE"
curl -s -o /dev/null -w '%{http_code}\n' -X POST localhost:3000/api/customers \
-H "Cookie: $COOKIE" -H 'content-type: application/json' -d '{"name":"Sneaky"}'
```

- [ ] The POST returns **403**

## 5. Form errors are never silent

A bug found during M1: a validation failure with no visible message meant the Save button appeared to do nothing.

- [ ] Paste 300 characters into any address field → a visible error appears
- [ ] On every form, an invalid field shows red text under **that** field
- [ ] No form ever fails with the button simply doing nothing

## 6. Audit

**Settings** → **Audit log** as admin.

- [ ] Everything above is recorded: branch created, customer created, tax rate added, payment method deactivated
- [ ] Entity types include [branch](#), [customer](#), [supplier](#), [tax\\_rate](#), [payment\\_method](#), [business](#)
- [ ] Expanding a row shows old → new values



## 7. Responsive

---

At ~375px, ~810px and full width:

- [ ] Customers, Suppliers and Branches lists become **cards** below 768px
- [ ] Business settings tabs scroll rather than wrap
- [ ] Forms stack to one column; buttons go full width
- [ ] Nothing scrolls sideways
- [ ] The sidebar's **Master data** and **Settings** groups appear in the mobile drawer

## What M1 deliberately does not include

---

- **No customer/supplier history tab yet.** Clicking a customer opens the edit form. The profile view showing their purchases, returns and credit arrives with M4/M5, when there is something to show. *This is a known gap against M1's stated criteria, carried into M2.*
  - No products, stock or IMEI — that is M2
  - Attachments have a working API and storage layer, but **no upload button on any screen yet**; the first screen that needs one is M3's purchase form
  - Logo upload is not wired to the business profile
- 

## M4 — Sales & Billing

**Delivers.** PRD FR-6.1 – FR-6.8, FR-26.1, FR-26.3, FR-26.4, FR-38.2, FR-38.3.

This is the screen the shop lives on. Everything before it existed so that this one could work: a bill is built, money is taken, stock drops, and an invoice comes out. Test it standing up, at speed, the way the counter will.

**Before you start.** You need stock. Run through M3 once to bring in at least one USED or ER handset with an IMEI, and one accessory with a quantity of ten.

### 1. Speed at the counter

---

Go to **Billing**.

- [ ] The search box already has focus — you can start typing without clicking
- [ ] Type an accessory name → results appear as you type
- [ ] With exactly one result, pressing **Enter** puts it on the bill
- [ ] Scan or paste an IMEI → that exact handset lands on the bill
- [ ] **Time yourself:** accessory → Cash → Save, keyboard only, under 20 seconds
- [ ] The whole flow needs no mouse

### 2. The bill

---

- [ ] A device line shows the IMEI or serial underneath the product name
- [ ] A device line shows its main type (USED / ER / ACT / GLOBAL)

- [ ] A GLOBAL device also shows **NEW CUT** where that applies
- [ ] A device line's quantity is fixed at 1 and cannot be edited
- [ ] An accessory line's quantity can be raised, and the total follows
- [ ] Adding the same IMEI twice is refused with a clear message
- [ ] Line discount reduces that line only; bill discount reduces the total
- [ ] **Staff** signed in cannot give a discount at all
- [ ] Totals: taxable + GST equals the total, exactly, to the paisa
- [ ] Change a quantity and the totals update instantly

### 3. The bill survives real life

---

- [ ] Build a half-finished bill and **reload the page** → it is still there
- [ ] Clear bill empties it, and asks nothing you would regret confirming
- [ ] Switch branch with a bill open → the bill clears (it belonged to the other branch's stock)

### 4. Payment

---

- [ ] **+ Cash** fills the full balance in one press
- [ ] Split across two methods (say Cash 500 + UPI rest) → both are recorded
- [ ] Pay less than the total with **no customer attached** → refused, because an unpaid balance needs someone to owe it
- [ ] Attach a customer, then save partly paid → the sale shows **PARTIAL**
- [ ] Overpaying is refused

### 5. Customer, without leaving the bill

---

- [ ] Press **New** beside the Customer box
- [ ] Name and phone only → **Add to bill**
- [ ] The new customer is created *and* attached, and the bill is untouched
- [ ] A duplicate phone number is refused with a message naming the clash
- [ ] Cancel closes the dialog and changes nothing

### 5c. Pickers do not silently truncate

---

Customers, suppliers and products are chosen through a **searchable picker**, never a fixed dropdown — a capped list drops records with no hint that it did.

- [ ] The Customer box on billing opens a search, not a long dropdown
- [ ] Typing a phone number finds the customer
- [ ] A customer created minutes ago is findable
- [ ] Same for **Supplier** on the purchase form

- ☐ Same for **Product** on the purchase form and on Add device

## 5b. Customer history (FR-6.7)

---

Open **Customers** and click a customer who has bought something.

- ☐ The **History** tab opens first — that is what you came to see
- ☐ Total spend, number of purchases, outstanding and last purchase date
- ☐ Every invoice is listed, newest first, with branch and item count
- ☐ Clicking an invoice number opens that sale
- ☐ Purchases from **every branch** appear, not just the current one
- ☐ A part-paid bill shows **PARTIAL**, and outstanding is non-zero and red
- ☐ A voided bill is listed but does **not** count towards spend or dues
- ☐ A customer who has bought nothing says so, rather than showing an empty table
- ☐ The **Details** tab still edits the customer and saves
- ☐ On a phone the history becomes cards and nothing overflows sideways

## 6. The invoice

---

Save a bill; you land on the sale.

- ☐ The invoice shows shop name, GSTIN, branch, invoice number and date
- ☐ Customer details appear, or "Walk-in customer"
- ☐ Every line shows quantity, price, tax and amount
- ☐ GST is broken out per rate
- ☐ **Print A4** produces a clean full-page invoice with no navigation on it
- ☐ **Print receipt (80 mm)** produces a narrow slip that fits the thermal roll
- ☐ **Download PDF** saves a file named after the invoice number
- ☐ Open the PDF: every figure matches the screen, and no character shows as a blank box
- ☐ The PDF's amounts are labelled INR (the rupee glyph is deliberately not used — see <src/server/pdf/invoice-pdf.tsx>)
- ☐ **Share** hands the PDF to the phone's share sheet (WhatsApp, mail); on a desktop browser it copies the invoice link instead

## 6b. The invoice is a statutory GST document (OQ-4)

---

First set **Settings** → **Business** → **GST state**, and a **GST state** on each branch. Give a few products an **HSN code** (8517 phones, 8544 cables).

- ☐ The invoice is headed **Tax Invoice**
- ☐ **Place of supply** appears, with the state code and name
- ☐ Each line shows its **HSN** code

- ☐ An ordinary local sale shows **CGST** and **SGST** as separate lines, each half the rate — never a single combined "GST 18%"
- ☐ CGST + SGST adds up to exactly the tax shown
- ☐ An **HSN summary** table appears below the totals on A4
- ☐ The HSN summary's taxable and tax columns add up to the invoice totals
- ☐ Set a customer's **GST state** to another state, bill them → the invoice shows a single **IGST** line and no CGST/SGST
- ☐ The customer pays the same total either way — only the split changes
- ☐ A product with no HSN still bills; the line shows "—" and stays out of the summary
- ☐ The downloaded PDF matches the screen on every one of these

## 7. Stock actually moved

---

- ☐ The handset you sold is now **Sold** on the device page
- ☐ Its IMEI no longer appears in billing search
- ☐ The accessory's branch stock dropped by the quantity sold
- ☐ The stock ledger shows one **SALE** row referencing the invoice

## 8. The other billing system (FR-38.2, FR-38.3)

---

- ☐ A **NEW** handset never appears in billing search at all
- ☐ Forcing one onto a bill is refused server-side too
- ☐ Open that device → **Sold in other system** is offered
- ☐ Mark it → status becomes **SOLD\_PENDING\_IMPORT** and stock drops now
- ☐ It disappears from billing search, so it cannot be sold twice
- ☐ An ECITY-channel device does **not** offer that button

## 9. Sales list

---

- ☐ Every saved sale is listed, newest first
- ☐ Search by invoice number, customer name or IMEI finds it
- ☐ Payment status filter: PAID / PARTIAL / UNPAID each return the right rows
- ☐ Branch filter narrows to that branch; a branch-scoped user sees only theirs
- ☐ From/To dates include both end days
- ☐ Filters survive a page reload (they are in the URL, so they are shareable)

## 10. Concurrency — needs two browsers

---

- ☐ Sign in on two windows, put the **same IMEI** on both bills
- ☐ Save both → the first succeeds, the second fails with a clear message

- [ ] The device is Sold exactly once, and only one sale exists
- [ ] Save a bill, then press Save again on a restored copy → still one sale

## 11. Responsive

---

- [ ] On a phone the billing screen is usable one-handed; nothing overflows sideways
- [ ] The cart and the totals panel stack rather than squeeze
- [ ] The invoice is readable on a phone
- [ ] On a large screen the search, cart and payment panel sit side by side

## What M4 deliberately does not include

---

- **No returns or exchanges** — that is M6
  - **No trade-in** on the bill — also M6
  - **No Excel import** of the other system's sales — that is M12; devices marked `SOLD_PENDING_IMPORT` are waiting for it
  - **Returns do not appear in customer history yet** — the tab says so; they arrive with M6. Payments now have their own Statement and Receipts tabs
  - No daily cash-up or Z-report — M7
- 

## Cross-cutting — Lists & pagination

Every list in the app is paginated server-side at **25 rows a page**. Check these on **Products, Devices, Customers, Suppliers, Purchases, Sales** and the **Audit log** — the control is the same one everywhere.

- [ ] Under each list: *"Showing 1–25 of N"*, with the real total
  - [ ] A list with more than 25 rows offers **Previous / Next**
  - [ ] A list with 25 or fewer offers no page buttons, but still states the count
  - [ ] On page 1, **Previous** is greyed out and does nothing when clicked
  - [ ] On the last page, **Next** is greyed out
  - [ ] **Next** advances, and the rows are genuinely different
  - [ ] Page 1's URL has no `?page=` on it — one address for the first page
  - [ ] Apply a filter, then page forward → **the filter is still applied**
  - [ ] Change a filter while on page 3 → you go back to page 1, not an empty page
  - [ ] Copy a page-2 URL into a new tab → same page, same filters
  - [ ] Browser back returns to the previous page of results
  - [ ] On a phone the buttons are big enough to tap and do not overflow sideways
- 

## M5 — Payments, Credit Sales & Customer Dues

**Delivers.** PRD FR-7.1 – FR-7.6.

Credit is where a shop quietly loses money. This module exists so that the answer to "who owes us what, and since when?" is one screen and always right.

**Before you start.** Set **Settings** → **Business** → **Default credit period** (30 days is the default). You need one product in stock and one customer.

## 1. A bill can leave the counter unpaid

---

- [ ] Bill something to a **named customer** and pay nothing → the **Credit terms** panel appears, with a due date already filled in
- [ ] Pay the bill in full → the credit panel disappears (a cash sale needs no due date)
- [ ] Pay part of it → the panel comes back
- [ ] Try to leave a balance with **no customer** → refused, because a walk-in cannot be given credit
- [ ] Change the due date by hand and save → the sale shows that date
- [ ] Save unpaid → the sale shows **UNPAID**; part-paid shows **PARTIAL**
- [ ] Open that sale: it shows **Outstanding**, the **due date**, your credit note, and a **Collect payment** button
- [ ] Past the due date, the sale says *overdue* in red
- [ ] A fully paid sale shows none of that panel

## 2. Customer dues (FR-7.6)

---

Open **Sell** → **Customer dues**.

- [ ] Your unpaid customer is listed, with the amount owed
- [ ] **Total outstanding** matches the sum of the rows
- [ ] The **By age** panel adds up to the same total
- [ ] A bill created today sits in the **0–7 days** column
- [ ] Search by customer name and by phone number
- [ ] Filter by branch → only that branch's bills count
- [ ] **Overdue only** hides a bill whose due date has not passed yet
- [ ] A customer who has paid everything drops off the list entirely
- [ ] On a phone the rows become cards and nothing scrolls sideways

## 3. Collecting money (FR-7.3)

---

- [ ] Press **Collect** on a row → the customer's open bills are listed, oldest first
- [ ] Type an amount → it is allocated down the list, oldest first, and you can see exactly which bills it lands on before saving
- [ ] **Settle everything** fills in the exact total owed
- [ ] **Choose invoices** lets you allocate by hand instead
- [ ] Allocating more than a bill owes is refused, and says which bill
- [ ] Save → you land on a printable receipt with its own number

- ☐ The bill is now **PAID** on the sales list *and* on the sale itself
- ☐ Three part payments settle a bill and flip it to Paid on the third

## 4. Paying at a different branch (FR-7.5)

---

- ☐ Bill something on credit at **Branch A**
- ☐ Collect the money at **Branch B** (change *Collected at* on the form)
- ☐ The receipt shows Branch B; the invoice still shows Branch A
- ☐ The statement shows both, each against its own line

## 4b. Branch-wise (FR-7.6)

---

The **By branch** panel appears on the dues screen when there is more than one branch with activity.

- ☐ Each branch shows outstanding, overdue, collected and receipt count
- ☐ The **All branches** row adds up the columns above it
- ☐ Outstanding here matches the total at the top of the page
- ☐ Bill on credit at Branch A, collect at Branch B → A's *outstanding* falls, B's *collected* rises. They are not supposed to match
- ☐ Change the **Collected from / to** dates → only the collected column moves; what is owed is owed regardless of the window
- ☐ Void a receipt → it stops counting as collected, and the debt returns to the branch that raised the bill

## 5. The receipt (FR-26.2)

---

- ☐ Shop name, branch, receipt number, date, customer
- ☐ The amount, the method, and the reference if you gave one
- ☐ **Applied to** lists each invoice and how much went to it
- ☐ Money beyond what was owed shows as **held on account as an advance**
- ☐ **Balance after this receipt** matches the customer's account
- ☐ Print A4 and Print receipt (80 mm) both produce clean output

## 6. Voiding a receipt

---

- ☐ **Void** is offered to the owner and manager
- ☐ It is **not** offered to counter staff — they may collect, not erase
- ☐ Voiding needs a reason
- ☐ After voiding: the receipt is marked VOIDED but still on record
- ☐ The invoice it settled is **UNPAID** again
- ☐ The customer's balance has gone back up
- ☐ The statement shows a **Reversal** line — the old line is untouched

- ☐ Voiding the same receipt twice is refused

## 7. The statement (FR-7.6)

---

Open a customer → **Statement**.

- ☐ Every movement, oldest first, with a running balance
- ☐ An invoice appears as a debit; a payment as a credit
- ☐ Clicking an invoice number opens that sale; a receipt opens that receipt
- ☐ The closing balance matches the figure at the top of the customer page
- ☐ A customer in credit shows a negative balance labelled *in credit*

## 8. The numbers reconcile

---

This is the one that matters. Pick any customer with dues:

- ☐ Add up what their open invoices still owe, by hand
- ☐ Compare with the balance on the dues list — they must be identical
- ☐ Compare with the statement's closing balance — also identical
- ☐ Void a receipt and repeat: all three still agree

## What M5 deliberately does not include

---

- **No returns or exchanges** — M6. A refund is not a negative payment
- **No cash drawer or daily closing** — M7. Money collected is recorded, but the day is not yet reconciled
- **No statements by email or WhatsApp** — M13
- **No interest, credit limits or dunning letters** — not in v1 at all

# M6 — Returns, Exchange & Trade-In

**Delivers.** PRD FR-8.1 – FR-8.5, FR-9.1 – FR-9.3, plus correcting a device.

Goods come back and old phones come in. The rule underneath everything here: **a returned handset is never immediately sellable again.**

**Before you start.** You need a completed sale with a handset on it and another with an accessory. Sign in as **admin** unless a step says otherwise.

## 1. Taking a return (FR-8.1)

---

- ☐ **Sell** → **Returns** → **Take a return** offers one search box
- ☐ Find a bill by its **invoice number**
- ☐ Find the same bill by **customer name**
- ☐ Find it by the **IMEI** of a handset on it



- [ ] Opening a sale and pressing **Take a return** gets you to the same place
- [ ] Only lines with something left to return are offered
- [ ] Returning 2 of 5 gives back exactly two fifths of that line
- [ ] Asking for more than remains is refused, and says how many are left
- [ ] Return 3 of 4, then try to return 2 more → refused, 1 left
- [ ] A whole bill returned at once is marked **FULL**; part of one is **PARTIAL**
- [ ] Goods can be returned to a **different branch** than they were sold at

## 2. A returned handset is not sellable (FR-8.2)

---

This is the one most likely to be got wrong. Test it properly.

- [ ] Return a handset → it appears in **Returns** → **Inspection queue**
- [ ] Search that IMEI at the till → **nothing comes up**
- [ ] The device page shows status **RETURNED**
- [ ] An accessory behaves differently: it goes straight back into branch stock and can be sold again immediately

## 3. Inspection (FR-8.3)

---

- [ ] **Inspect** offers Available, Used, Damaged, Repair required
- [ ] Grading **Available** → the handset leaves the queue and reaches the till
- [ ] Grading **Used** → also sellable, and the condition is recorded
- [ ] Grading **Damaged** → status DAMAGED, still not sellable
- [ ] Grading **Repair required** → status REPAIR, still not sellable
- [ ] Notes typed during inspection appear on the device
- [ ] Inspecting something that is not awaiting inspection is refused
- [ ] Signed in as **staff**: the queue is visible but there is **no Inspect button** — releasing stock for sale is not the counter's decision

## 4. Classification survives the whole path (FR-8.4)

---

The acceptance test for this module. Do it exactly.

- [ ] Buy a handset as **GLOBAL** with **NEW CUT** ticked
- [ ] Sell it, then return it
- [ ] In the inspection queue it still shows **GLOBAL**
- [ ] Grade it **Used**
- [ ] On the device page it **still reads GLOBAL and NEW CUT**, with condition Used recorded separately

Condition and classification are different facts. Grading a handset's condition must never rewrite what kind of device it is.

## 5. The money (FR-8.5)

- ☐ Refund to a **payment method** — money goes back out
- ☐ Refund to the **customer's account** — no cash moves, they owe less
- ☐ A walk-in cannot have their account credited; it is refused
- ☐ A **deduction** (restocking fee) reduces what is handed back
- ☐ **The one that matters:** a customer who paid in full, returned the goods and got cash back ends with a **zero balance** — not in credit. Check Customers → their Statement
- ☐ Returning against an **unpaid** bill just reduces what they owe

## 6. Trade-in (FR-9.1 – FR-9.3)

On the billing screen, with something in the cart:

- ☐ **Trade-in** → **Add** takes IMEI, type, storage, colour, battery, condition notes, estimated value and agreed value
- ☐ Accepting it adds the old handset to **this branch's stock** immediately
- ☐ Its device page shows the type you chose, including GLOBAL/NEW CUT
- ☐ Its history starts with a **PURCHASED** event
- ☐ Its purchase price is the **agreed value**
- ☐ It is marked as sold here, never the other billing system
- ☐ On the bill: **Bill total** stays the full price, and the trade-in shows as a deduction against what is still to pay
- ☐ Save the bill → the difference is what the customer owed
- ☐ The invoice still shows the full price and full GST — a trade-in is not a discount, and the trade-in appears under the total as settlement, beside the cash, with the old handset's IMEI
- ☐ **Take one in for the full price of the phone and pay nothing.** The bill saves, reads **PAID**, and shows no outstanding balance and no due date. *A bill settled with the handset must not leave a receivable to chase*
- ☐ Do the same for a **named customer**: their statement shows the sale, the trade-in against it, and a balance of zero
- ☐ The sale list and the customer dues screen both read that bill as PAID — settlement is answered in one place, so they cannot disagree
- ☐ Trading in with **no customer** on the bill is allowed when the handset covers the whole total; it is refused only when something is left owing

## 6b. Refunding no more than was taken

- ☐ Sell on **credit** to a customer, pay nothing, then return the goods and choose to refund to **Cash** → the refund is ₹0 and their balance goes to zero. *They never paid, so nothing goes back out*
- ☐ Sell ₹1,000, pay ₹400, return everything, refund to Cash → ₹400 goes back and the remaining ₹600 comes off their account
- ☐ Same unpaid credit sale, but refund to the **customer's account** → the full value is credited. *A credit note is not money leaving the till*
- ☐ Return the rest of a part-returned bill → the two refunds together never exceed what was paid

## 6c. The GST switch (FR-2.6, FR-26.6)

**Settings** → **Business** → **GST**. Turn *Registered for GST* off and save.

- ☐ The **GST number** and **GST state** fields disappear from business settings, and so does the **Tax** tab and **Prices include tax**
- ☐ A new product form has no **HSN code** and no **Tax rate**
- ☐ A new customer, supplier and branch form has no **GST number**
- ☐ A device form has no **Tax rate**
- ☐ At the till, the total shows **Bill total** only — no Taxable, no Tax
- ☐ Sell a phone at ₹10,000 and take ₹10,000 → the bill total is exactly ₹10,000. *The customer pays the price on the label*
- ☐ Its invoice is headed **Invoice**, not Tax Invoice, and shows no GSTIN, no HSN column, no tax column and no HSN summary
- ☐ Download the PDF — same on paper as on screen

**Now turn GST back on and save.**

- ☐ Reopen that bill. It **still** reads Invoice, with no GSTIN and no tax. *This is the important one: nothing stores the PDF, so every reprint is re-rendered. A bill issued before the shop registered must never reprint as a tax invoice*
- ☐ Sell another phone → this one **is** a Tax Invoice, with GSTIN, HSN and the tax breakdown
- ☐ Turn GST off again and reopen that taxed bill → it still shows its GST. *It works in both directions*
- ☐ Turn GST back on before carrying on with the rest of the checklist

## 7. Correcting a device

- ☐ A device page shows **Edit** for admin and manager
- ☐ Signed in as **staff** there is no Edit button
- ☐ Change the **main type** and save
- ☐ The device history shows a **RECLASSIFIED** entry with what changed — corrections are recorded, not silent
- ☐ Change **Billed in** from the other system to ECITY → that handset now appears at the till. *This is the fix for NEW stock that was invisible*
- ☐ NEW CUT on anything that is not GLOBAL is refused
- ☐ A **sold** device cannot be edited — an issued invoice describes it
- ☐ The IMEI is shown but not editable

## What M6 deliberately does not include

- **No voiding a return.** Take a new sale if goods go back out
- **No repair workflow** — REPAIR is a status, not a job card
- **No automatic valuation** of trade-ins; the shop decides the number
- **No returns against the other billing system's sales** — those bills do not exist here until M12 imports them

## Housekeeping

The end-to-end tests create real branches and customers in your development database. After a test run you will see extra rows. To get back to a clean state:

```
dropdb ecity && createdb ecity
psql -d ecity -c "CREATE ROLE ecity WITH LOGIN PASSWORD 'ecity' SUPERUSER;"
npm run db:migrate && npm run db:seed
```

## Sign-off

| Tested by      |             |
|----------------|-------------|
| Date           |             |
| Build / commit |             |
| Result         | pass / fail |
| Issues raised  |             |

## M7 — Expenses, Cash Drawer, Accounts & Daily Closing

Sign in as **admin** unless a step says otherwise. Work a whole day through in one sitting — the point of this module is that the day balances at the end.

### 1. The drawer sees every rupee (FR-11.2)

Open **Cash drawer** and note *Expected in the till*. After each step below, come back and check it moved the right way and by the right amount.

- [ ] Bill something for **cash** → expected goes **up**
- [ ] Bill something by **UPI** → expected does **not** move. *UPI never touches the till*
- [ ] Collect a customer's outstanding balance in **cash** → **up**
- [ ] Record a **cash expense** → **down**
- [ ] Pay a supplier in **cash** → **down**
- [ ] Refund a return in **cash** → **down**
- [ ] Every one of those appears as its own line under the total, with a time
- [ ] Switch branch → the other branch's till is a separate figure. *One branch's takings must never appear in another's*

### 2. Expenses (FR-10.1 – FR-10.3)

- [ ] **Record an expense** takes branch, category, amount, date, how it was paid, description and reference
- [ ] Choosing a non-cash method offers an **account**; choosing cash does not

- [ ] It appears in the list, newest first, with who recorded it
- [ ] Click the category → the expense opens on its own page
- [ ] **Attach a receipt** — a photo or PDF — and it shows on the expense
- [ ] A voided expense offers no upload
- [ ] **Total spent** is over everything in the filter, not just this page
- [ ] **Void** one → it asks why, and refuses an empty reason
- [ ] After voiding, the money is back in the drawer and the expense is gone from the live list. *It is voided, not deleted — tick "include voided" and it is still there with its reason*

### 3. Accounts (FR-12.1 – FR-12.4)

---

- [ ] **Add account** for a bank and one for UPI, with an opening balance
- [ ] An account can belong to one branch or to **all branches**
- [ ] **Transfer** between them → both balances move, by the same amount
- [ ] Click an account name → its **ledger**: every movement, newest first, with a running balance you can prove the total from
- [ ] The two legs of the transfer appear, one in each account
- [ ] **Reconcile** one, entering a statement balance ₹500 *below* the books
- [ ] The balance does **not** change to match. The ₹500 shows as **unreconciled**. *A real difference wants an explanation, not an automatic adjustment*

### 4. Closing the day (FR-13.1 – FR-13.4)

---

- [ ] **Daily closing** shows sales, credit given, collected, returns and refunds, what each payment method took, and the expenses by category
- [ ] It shows **invoices and items**. Sell three of one thing on one bill → one invoice, three items
- [ ] Type a counted amount ₹**200 below** expected → it says ₹**200 short** as you type, before you commit to anything
- [ ] Close the day → the shortage is recorded with the branch, your name and the time
- [ ] The drawer for that day now reads **CLOSED**

### 5. A closed day is never rewritten (OQ-5) — *the important one*

---

- [ ] Signed in as **staff**, try to record an expense dated into the closed day → refused
- [ ] As **admin**, record one → allowed, and the movement is tagged **After close**
- [ ] Go back to the closing screen. It says **this day was corrected after it was closed**, and shows both figures: what was signed, and what the movements now come to
- [ ] The signed **expected, counted and difference are unchanged**. *This is the whole decision. If a closing could be edited, coming up short today and adjusting yesterday would make the difference disappear*

## 6. Reopening a day

---

- [ ] On a closed day, **Reopen this day** → asks why, and refuses an empty reason
- [ ] It reopens, and the day can be closed again
- [ ] Close a **later** day for the same branch, then try to reopen the earlier one → **refused**, and it tells you to post a correction instead
- [ ] **Closing history** shows the reopened day marked **Reopened**. *The voided closing is kept — that a day was closed and reopened is part of the record*

## 7. Two branches (FR-13.5)

---

- [ ] Close each branch's day separately
- [ ] **All branches** shows both side by side and the consolidated totals add up
- [ ] A branch that has not closed still appears, marked **Open**, with what it should have

## 8. Tomorrow starts from what was counted

---

- [ ] Close a day counting ₹100 **less** than expected
- [ ] Open the next day's drawer → its **opening** is what you counted, not what was expected. *The drawer starts with the cash actually in it, so a shortage does not repeat itself every morning*

## 9. Correcting a purchase (carried in from M5)

---

- [ ] A purchase page shows **Correct details** for admin and manager
- [ ] It offers the **supplier bill number, date and notes** — and nothing else
- [ ] There is **no** way to change a line, quantity or cost. *Stock has moved and the supplier ledger has been posted; reversal is the honest tool*
- [ ] Save a corrected bill number → it shows on the purchase
- [ ] Signed in as **staff**, there is no Correct details button

## What M7 does not do

---

- **No cash counting by denomination**; one counted total per day
- **Legacy sales are not in the expected figure**. They take cash into the same physical till, so until M12 imports that day's file, expected cash is short by whatever the other system took. The closing screen is built to be gated on that import, with a recorded override — the gate is wired in M12

## M8 — Branch Transfers & Stock Adjustments

You need two branches with stock. Sign in as **admin** unless a step says otherwise.

## 1. Requesting (FR-3.6)

---

- [ ] **Transfers** → **Request a transfer**
- [ ] Picking a *From* branch shows only what **that branch actually holds** — handsets in stock and accessories with a count
- [ ] Changing the *From* branch clears what you had added. *Those items were on the other branch's shelf*
- [ ] A handset adds as one line with its IMEI; an accessory lets you type a quantity, capped at what is on hand
- [ ] Save → it appears as **REQUESTED**
- [ ] Nothing has moved yet: the handset is still sellable at the sending branch

## 2. Approving and sending

---

- [ ] The transfer list has **Waiting for approval** and **To receive** — the approval queue and the receiving screen, one click away
- [ ] The transfer offers **Approve**, then **Dispatch** — not both at once
- [ ] Signed in as **staff**, a requested transfer offers no Approve button. *Sending stock out of a branch is a manager's decision*
- [ ] As a manager assigned to the **receiving** branch only, the transfer offers no Approve or Dispatch — it says the other branch sends it
- [ ] Once in transit, that same manager **can** receive it, and a manager at the **sending** branch cannot. *Signing for a delivery is the receiving branch's job — whoever packed the box cannot attest it arrived*
- [ ] After **Dispatch** it reads **IN TRANSIT**

## 3. In transit belongs to nobody — the important one

---

- [ ] At the **sending** branch, search the IMEI at the till → **not offered**
- [ ] Switch to the **receiving** branch and search it → **not offered**
- [ ] The accessory count at the sending branch has **gone down** by what was sent
- [ ] The receiving branch's count has **not gone up** yet. *It is in a van. If it were sellable at both ends it would be sold twice*

## 4. Receiving (FR-3.7)

---

- [ ] The receiving screen offers a **scan box**, focused and ready
- [ ] Nothing is ticked to begin with. *A box you scan into is safer than one you untick — a handset the scanner misses shows as short instead of passing quietly*
- [ ] Scan (or type and press Enter) each IMEI → its line ticks
- [ ] A handset that will not scan can still be ticked by hand
- [ ] **Receive in full** → the transfer reads **RECEIVED**
- [ ] The exact IMEI is now at the receiving branch and sellable there
- [ ] Its device history shows **TRANSFERRED OUT** and **TRANSFERRED IN**, each naming **both branches** and the date. *Not one branch with the other hidden in the detail — M9's timeline reads these directly*

- [ ] Its **main type** and **NEW CUT** are unchanged. *A journey is not a reclassification*
- [ ] The accessory count at the receiving branch has gone up

## 5. Something did not arrive

---

- [ ] Send another handset and, on receipt, **untick** it
- [ ] A warning appears explaining it will be marked lost, and asks what happened
- [ ] After receiving, the transfer is flagged **Short on receipt** with your note
- [ ] The handset reads **LOST**, not still in transit. *It left one branch and reached no other — it has to be accounted for, not stranded*
- [ ] Try a partly received accessory line: send 5, receive 3 → only **3** arrive and the transfer is flagged short

## 5b. Something arrived that should not have

---

- [ ] On a receiving screen, scan an IMEI that is **not on the transfer**
- [ ] It is listed as unexpected, with a warning, and can be removed if you mis-scanned
- [ ] Receive → the transfer is flagged, and the stray IMEI is named in the discrepancy note
- [ ] That handset has **not** been moved to this branch. *Recording it is right; moving it would be inventing a transfer nobody authorised*

## 6. Cancelling

---

- [ ] Cancel a **requested** transfer → nothing moves anywhere
- [ ] Cancel one **in transit** → it asks why, and refuses an empty reason
- [ ] The handset is back at the **sending** branch and sellable again
- [ ] Its history shows the return **without** claiming it came from the other branch. *It never got there*
- [ ] Accessory stock is back at the sending branch too
- [ ] A **received** transfer offers no Cancel. *A completed movement is corrected by transferring back, not by rewinding it*

## 7. Adjustments (FR-28.1 – FR-28.3)

---

- [ ] **Adjustments** → **Adjust stock** asks first whether it is an accessory count or one handset
- [ ] For an accessory: pick it, type what you **actually counted**, and the change is shown as you type
- [ ] Entering the number the system already has is refused. *There is nothing to correct*
- [ ] Save with reason **Miscount** → the count changes and the entry shows **10 → 7** with who and when
- [ ] For a handset: **Miscount is not offered** — only Damage and Loss, and it explains that a wrong record is a device correction instead
- [ ] Adjust a handset as **Damage** → it leaves sellable stock and the till no longer offers it
- [ ] The adjustment lists the handset's **main type** and **NEW CUT as they were at the time**
- [ ] Click an adjustment → it opens on its own page



- [ ] **Attach evidence** — a photo of the damage or the count sheet
- [ ] The page says an adjustment is never edited, and that a wrong one is corrected by a second
- [ ] Signed in as **staff**, there is no Adjust stock button

## 8. Navigation

---

- [ ] **Transfers** and **Adjustments** appear under Inventory
- [ ] The transfer list filters by stage — **Requested** is the approval queue, **In transit** is the receiving one

## What M8 does not do

---

- **No printed transfer note** to travel with the goods
- **No partial dispatch**; a transfer goes in one movement or not at all
- **No printed barcode labels**; a scanner types the IMEI into the scan box like a keyboard, which is how these scanners work anyway
- **No transfer between businesses**, only between branches of one